

CUSTOMER BEHAVIOR

--Q1. What is the total revenue generated by male vs. female customers?

```
select gender, SUM(purchased_amount) as revenue  
  
from customer  
  
group by gender
```

--Q2. Which customers used a discount but still spent more than the average purchase amount?

```
select customer_id, purchased_amount  
  
from customer  
  
where discount_applied = 'Yes' and purchased_amount >= (select  
AVG(purchased_amount) from customer)
```

-- Q3. Which are the top 5 products with the highest average review rating?

```
select item_purchased, round(avg(review_rating::numeric),2) as "Average Product  
Rating"  
  
from customer  
  
group by item_purchased  
  
order by avg(review_rating) desc  
  
limit 5
```

--Q4. Compare the average Purchase Amounts between Standard and Express Shipping.

```
select shipping_type,  
  
ROUND(AVG(purchased_amount),2)  
  
from customer  
  
where shipping_type in ('Standard','Express')
```

group by shipping_type;

--Q5. Do subscribed customers spend more? Compare average spend and total revenue

--between subscribers and non-subscribers.

```
SELECT subscription_status,
       COUNT(customer_id) AS total_customers,
       ROUND(AVG(purchased_amount),2) AS avg_spend,
       ROUND(SUM(purchased_amount),2) AS total_revenue
FROM customer
GROUP BY subscription_status
ORDER BY total_revenue,avg_spend DESC;
```

--Q6. Which 5 products have the highest percentage of purchases with discounts applied?

```
SELECT item_purchased,
       ROUND(100.0 * SUM(CASE WHEN discount_applied = 'Yes' THEN 1 ELSE 0
END)/COUNT(*),2) AS discount_rate
FROM customer
GROUP BY item_purchased
ORDER BY discount_rate DESC
LIMIT 5;
```

--Q7. Segment customers into New, Returning, and Loyal based on their total number of previous purchases, and show the count of each segment.

with customer_type as (

```
SELECT customer_id, previous_purchases,
CASE
```

```
    WHEN previous_purchases = 1 THEN 'New'
    WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'
    ELSE 'Loyal'
  END AS customer_segment
FROM customer)
```

```
select customer_segment,count(*) AS "Number of Customers"
from customer_type
group by customer_segment;
```

--Q8. What are the top 3 most purchased products within each category?

```
WITH item_counts AS (
    SELECT category,
           item_purchased,
           COUNT(customer_id) AS total_orders,
           ROW_NUMBER() OVER (PARTITION BY category ORDER BY COUNT(customer_id)
                               DESC) AS item_rank
    FROM customer
    GROUP BY category, item_purchased
)
SELECT item_rank,category, item_purchased, total_orders
FROM item_counts
WHERE item_rank <=3;
```

--Q9. Are customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?

```
SELECT subscription_status,
       COUNT(customer_id) AS repeat_buyers
FROM customer
```

```
WHERE previous_purchases > 5
```

```
GROUP BY subscription_status;
```

--Q10. What is the revenue contribution of each age group?

```
SELECT
```

```
    age_group,
```

```
    SUM(purchased_amount) AS total_revenue
```

```
FROM customer
```

```
GROUP BY age_group
```

```
ORDER BY total_revenue desc;
```